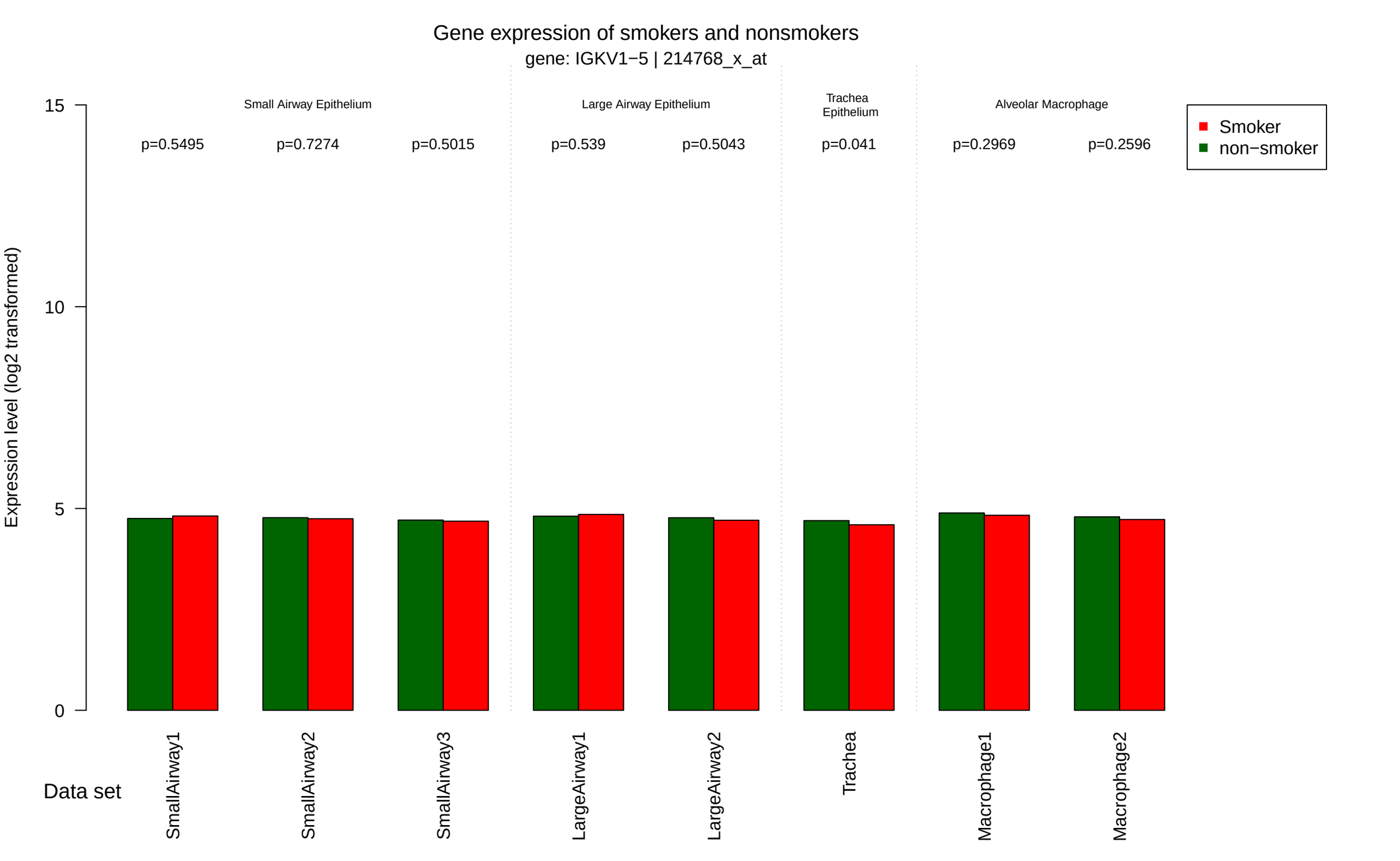
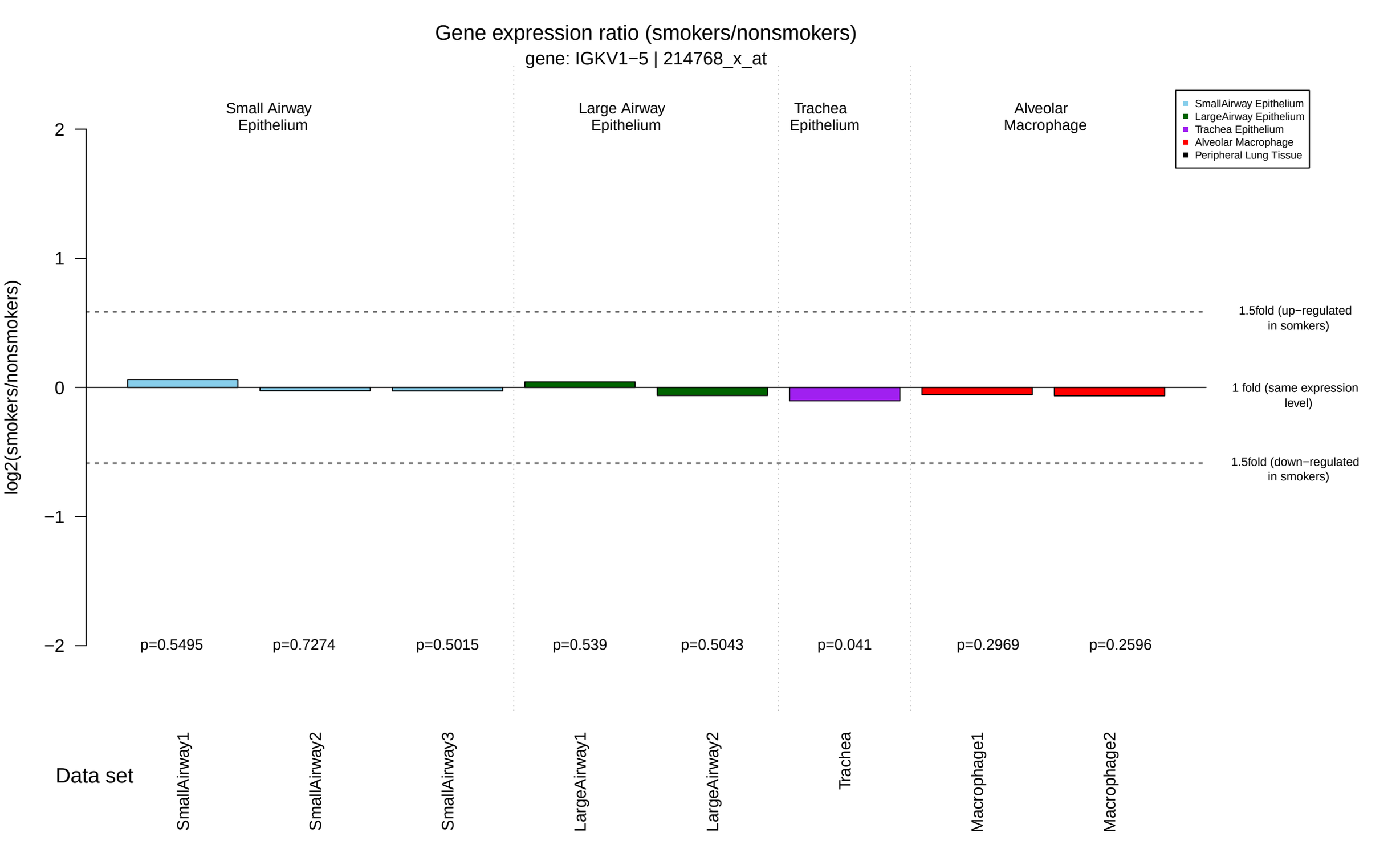
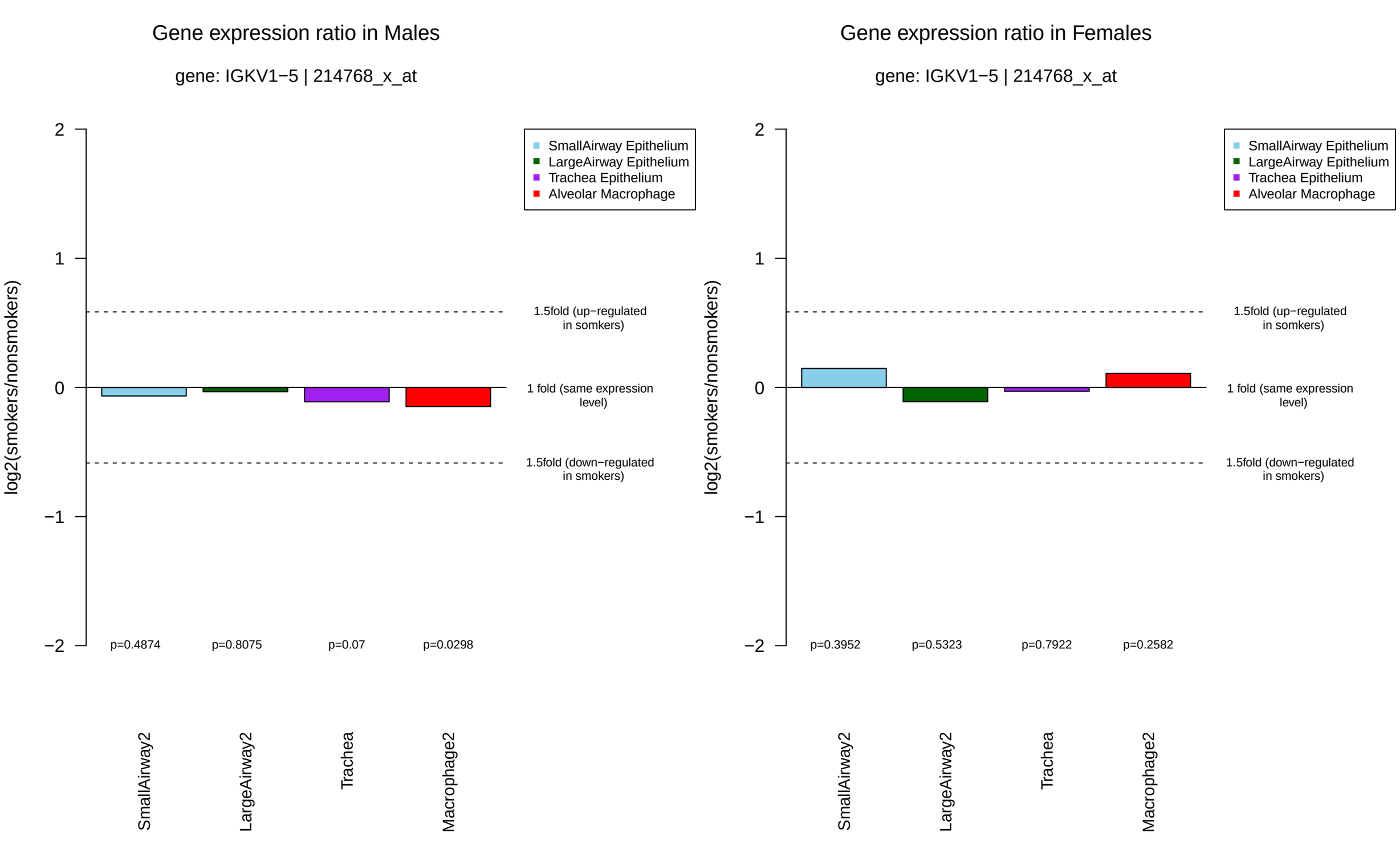
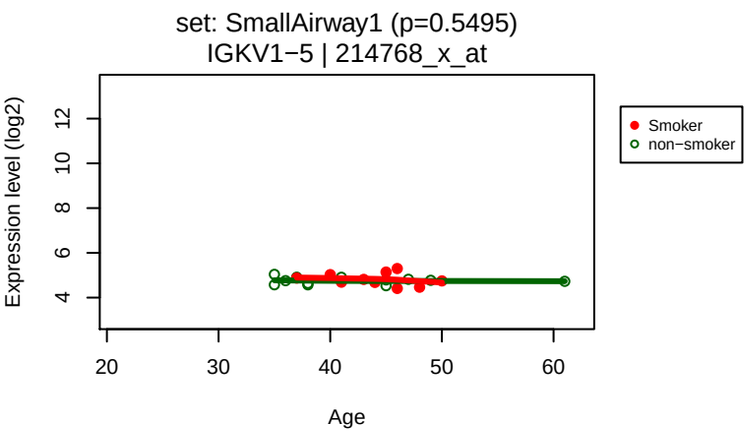
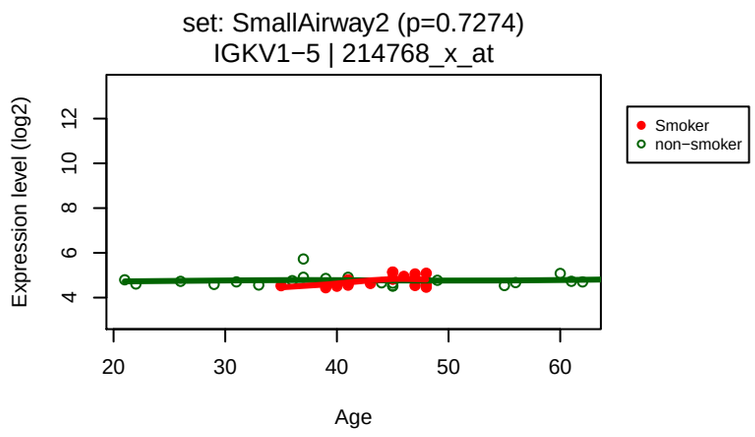
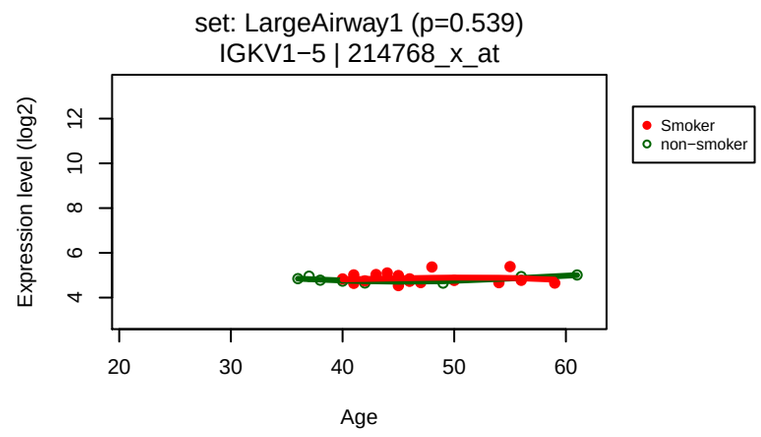

Gene expression of smokers and nonsmokers
gene: IGKV1−5 | 214768_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5495
p=0.7274
p=0.5015
p=0.539
p=0.5043
p=0.041
p=0.2969
p=0.2596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

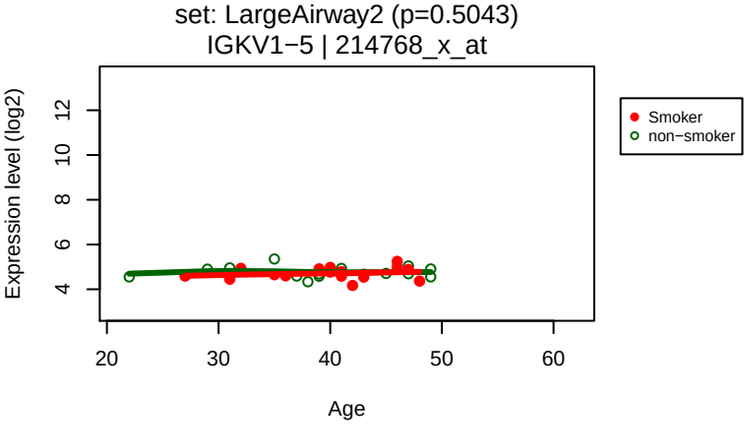

Gene expression ratio (smokers/nonsmokers)
gene: IGKV1−5 | 214768_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5495
p=0.7274
p=0.5015
p=0.539
p=0.5043
p=0.041
p=0.2969
p=0.2596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

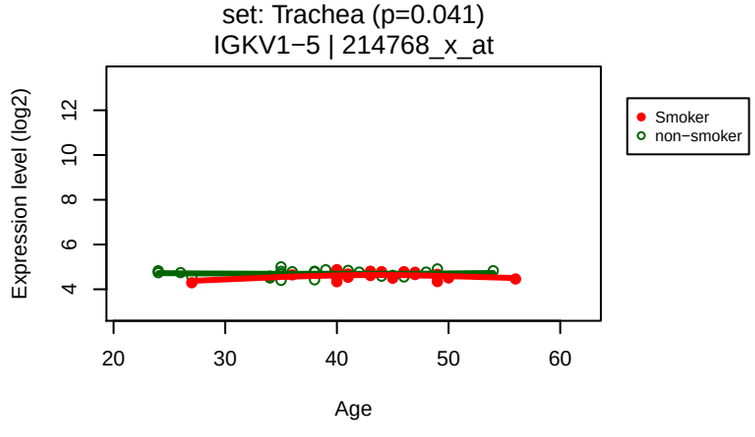

Gene expression ratio in Males
gene: IGKV1−5 | 214768_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4874
p=0.8075
p=0.07
p=0.0298
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGKV1−5 | 214768_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3952
p=0.5323
p=0.7922
p=0.2582
SmallAirway2
LargeAirway2
Trachea
Macrophage2

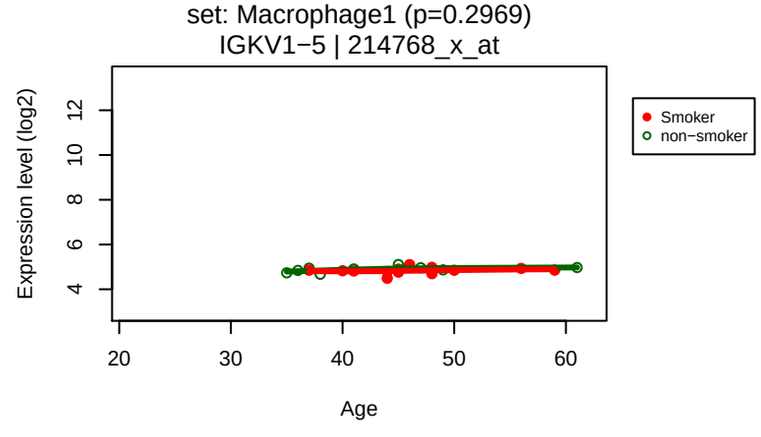

set: SmallAirway1 (p=0.5495)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker

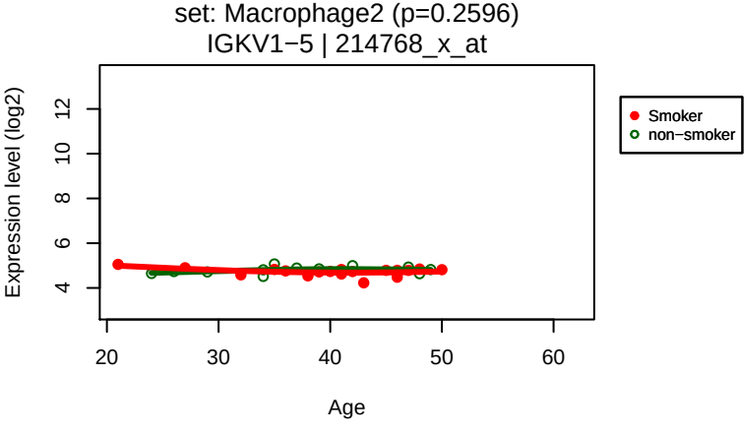

set: SmallAirway2 (p=0.7274)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker

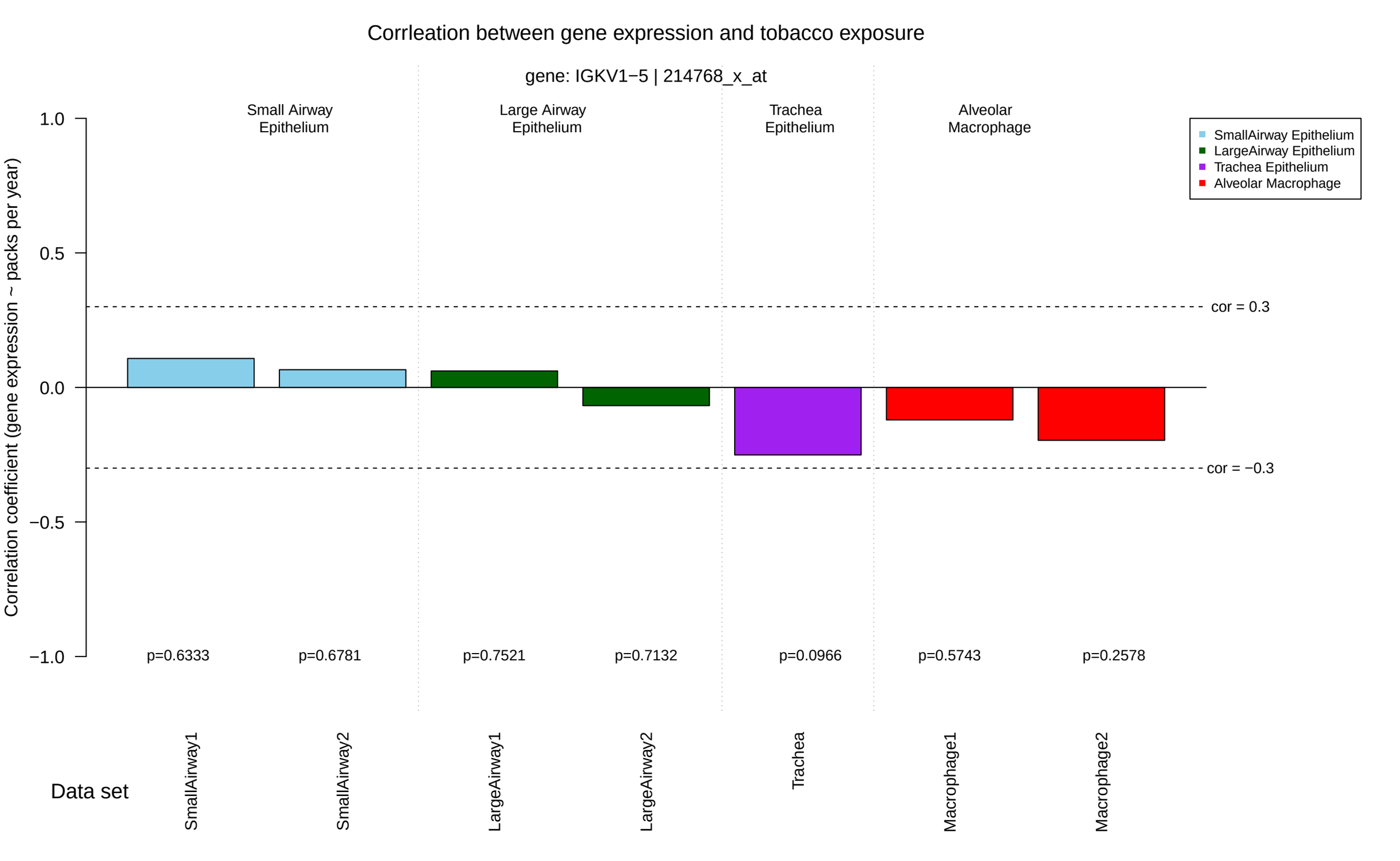

set: LargeAirway1 (p=0.539)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5043)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.041)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2969)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2596)
IGKV1−5 | 214768_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGKV1−5 | 214768_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6333
p=0.6781
p=0.7521
p=0.7132
p=0.0966
p=0.5743
p=0.2578
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2